

Steam Reviews Dataset: A Big Data Analysis

by Ali Daoud

Abstract

This project presents a Big Data analysis of the Steam Reviews Dataset using Apache Hadoop and Apache Spark. The dataset was obtained from Kaggle and contains more than 40 million user reviews from the Steam gaming platform, with a total size of approximately 7.6 GB. Due to its large volume and unprocessed nature, the dataset was suitable for distributed storage and large-scale analytics.

The data was uploaded to the Hadoop Distributed File System (HDFS) and processed using PySpark. Several preprocessing steps were performed, including schema inspection, filtering of invalid records, handling missing values, and conversion of selected columns into numerical formats. Descriptive and exploratory analyses were then conducted to identify patterns related to player recommendations, game popularity, language distribution, purchase behavior, and player engagement.

The results showed that positive reviews significantly outnumbered negative reviews. English was the most common review language, and popular multiplayer games generated the largest number of reviews. Additional analysis revealed relationships between recommendation status, purchase behavior, helpful votes, and player playtime. The project demonstrates how Hadoop and Spark can be used to efficiently process large datasets and extract meaningful insights from gaming-related data.

Innehåll

Steam Reviews Dataset: A Big Data Analysis by Ali Daoud	1
1.1 Dataset	2
1.2 Introduction to the Tools.....	2
1.2.1 Hadoop.....	2
1.2.2 Spark	2
2. Materials and Methods.....	3
2.1 Hardware.....	3
2.2 Data Collection.....	3
2.3 Data Pre-processing	3
2.4 Data Analysis, Representation and Visualization	3
2.5 Security	3
3. Results.....	4
4. Discussion.....	7
5. Conclusion.....	7
6. References.....	7
7.Appendix	7

1. Introduction

The rapid growth of digital gaming platforms has generated massive amounts of user-generated content in the form of online reviews. These reviews provide valuable information about player satisfaction, game quality, and consumer behavior. As gaming platforms continue to expand, the volume of available review data becomes increasingly difficult to process using traditional analytical methods. Big Data Analytics offers an effective approach for handling large-scale datasets and extracting meaningful insights from millions of records.

This project focuses on the Steam Reviews Dataset obtained from Kaggle. The dataset contains more than 40 million game reviews and occupies approximately 7.6 GB of storage, making it suitable for Big Data Analytics. The dataset includes information about games, user recommendations, languages, purchase status, playtime statistics, and review metadata. Due to its size and complexity, the dataset represents a realistic big data scenario requiring distributed storage and processing technologies.

The main objective of this project is to analyze player reviews and identify patterns related to game popularity, recommendation behavior, language distribution, purchase behavior, and player engagement. Hadoop Distributed File System (HDFS) was used for distributed storage of the dataset, while Apache Spark and PySpark were used for large-scale data processing and analysis. The project demonstrates how modern big data technologies can be applied to extract valuable insights from large gaming datasets.

1.1 Dataset

The dataset used in this project is the Steam Reviews Dataset downloaded from Kaggle. The dataset contains approximately 40.8 million reviews collected from the Steam gaming platform. Each record contains information such as game identifier, game name, review text, review language, recommendation status, helpful votes, purchase information, and player activity statistics. The raw dataset size is approximately 7.6 GB and therefore qualifies as a big data dataset according to the requirements of the project.

Dataset Summary

Metric	Value
Total reviews	21,747,371
Positive reviews	19,022,610
Negative reviews	2,724,761
Number of games	315
HDFS dataset size	7.6 GB

1.2 Introduction to the Tools

1.2.1 Hadoop

Apache Hadoop was used in this project as the storage platform for the Steam Reviews Dataset. Since the dataset contains over 40 million reviews and occupies approximately 7.6 GB, storing and managing the data in a distributed environment was appropriate. The dataset was uploaded to HDFS, where it was stored as distributed blocks that could be accessed efficiently by Apache Spark during the analysis process. This setup enabled reliable handling of large amounts of data and supported scalable processing throughout the project.

1.2.2 Spark

Apache Spark was used as the main processing framework for analyzing the Steam Reviews Dataset. Through PySpark, large amounts of review data could be loaded directly from HDFS and processed efficiently using distributed computations. Spark DataFrames were used to perform filtering, aggregation, sorting, and statistical calculations on millions of records. The framework was also used to generate the results presented

in this report, including analyses of recommendation behavior, language distribution, game popularity, purchase patterns, and player engagement.

2. Materials and Methods

2.1 Hardware

Table 1. Hardware specifications used in the project.

Component	Specification
CPU	Intel Core i5-1135G7 (11th Gen, 4 Cores, 8 Threads)
RAM	8 GB
Storage	SSD Storage
Operating System	Windows 11 Enterprise
Hadoop Version	Hadoop 3.3.6
Spark Version	Spark 3.5.1

2.2 Data Collection

The Steam Reviews Dataset was downloaded from Kaggle and transferred to the Hadoop Distributed File System. The dataset was uploaded to HDFS using Hadoop commands and stored in the directory `"/data/steam_reviews.csv"`. After successful storage verification, Apache Spark was connected to HDFS and used to access the dataset for analysis.

2.3 Data Pre-processing

The dataset was initially loaded into PySpark DataFrames using Spark's CSV reader with header support enabled. During the preprocessing phase, the structure of the dataset was inspected through schema analysis. Several columns containing numerical information were initially stored as string values and therefore required type conversion before statistical operations could be performed. Missing values and invalid records were identified during exploratory analysis. Language records containing null values were excluded from language-based analysis to improve result accuracy. Columns such as `votes_helpful` and `author.playtime_forever` were converted into numeric formats to enable aggregation functions and statistical calculations. This preprocessing stage ensured that the dataset was suitable for further analysis.

2.4 Data Analysis, Representation and Visualization

The analysis focused on identifying patterns in user recommendations, game popularity, language distribution, purchase behavior, and player engagement. Descriptive statistical techniques were applied using Spark DataFrame operations such as grouping, counting, aggregation, sorting, and filtering. Several analytical queries were executed to determine the distribution of positive and negative recommendations, identify the most reviewed games, analyze language usage among players, investigate purchase behavior, and compare average helpful votes and average playtime between different review categories. The results were visualized using bar charts and tables to facilitate interpretation and communication of findings.

2.5 Security

In this project, Hadoop audit logging was enabled to monitor activity within HDFS during data storage and processing. The logs recorded file access and system operations, making it possible to verify that the dataset was uploaded and accessed correctly throughout the analysis. This functionality provided basic traceability and helped identify potential issues during data processing. Although the project did not involve sensitive

personal information, logging demonstrated how monitoring mechanisms can support accountability and system management in large-scale data environments.

3. Results

The Steam Reviews Dataset contains more than 40 million reviews and provides valuable insights into player behavior and game popularity. Figure 1 presents the distribution of recommendations among users. The results show that positive recommendations significantly outnumber negative recommendations, indicating that most players were satisfied with their gaming experiences.

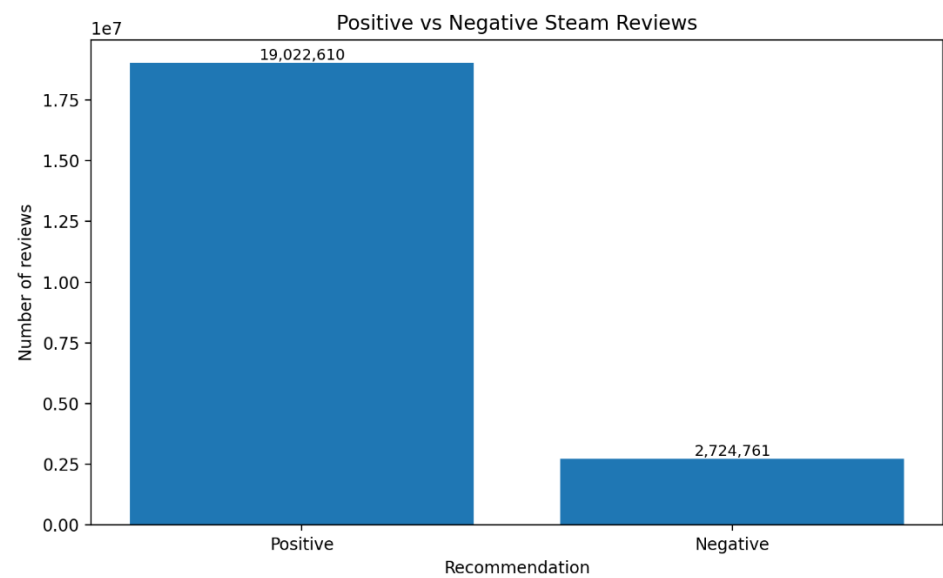


Figure 2 presents the ten most reviewed games in the dataset. PLAYERUNKNOWN'S BATTLEGROUNDS received the largest number of reviews, followed by Grand Theft Auto V and Tom Clancy's Rainbow Six Siege. These results highlight the popularity and engagement levels of major multiplayer games.

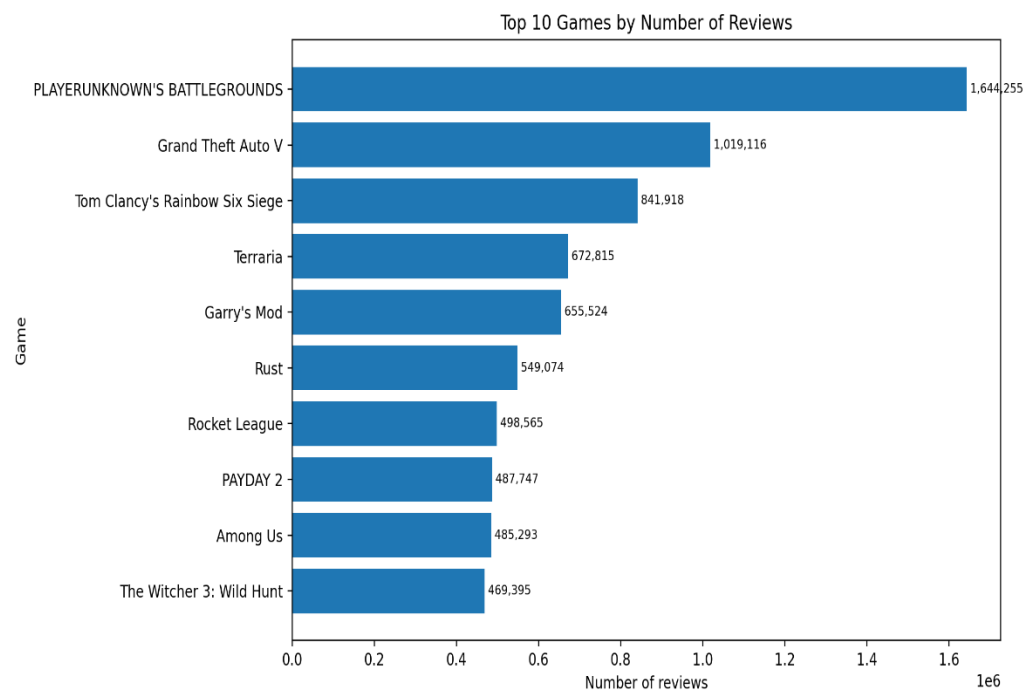


Figure 3 illustrates the distribution of review languages. English was by far the most frequently used language, followed by Simplified Chinese and Russian. This reflects the global nature of the Steam platform and its diverse user base.

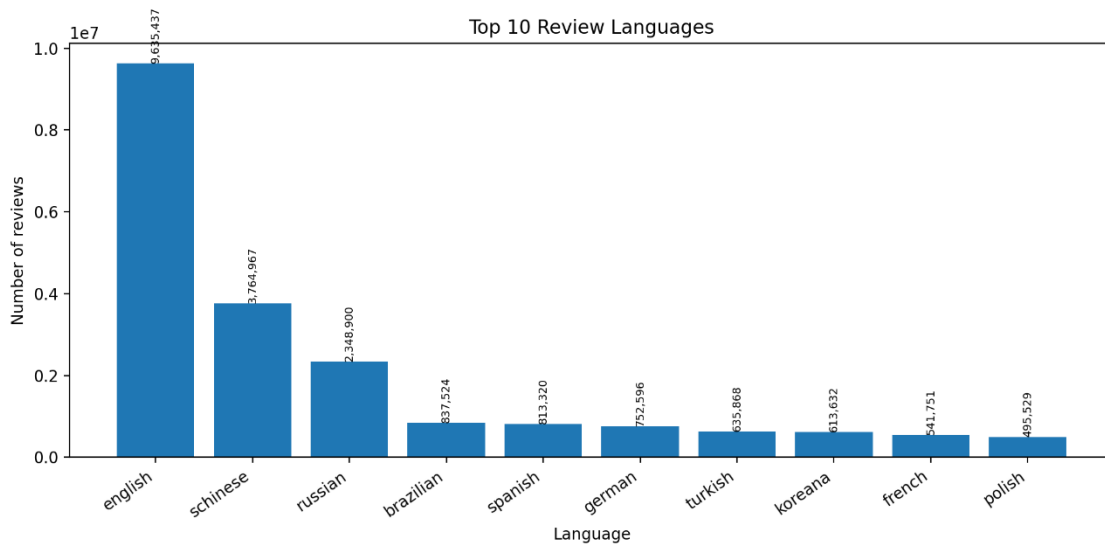


Figure 4 compares recommendation status with purchase behavior. Most positive recommendations originated from users who had purchased the game through Steam, suggesting a strong relationship between ownership and positive engagement.

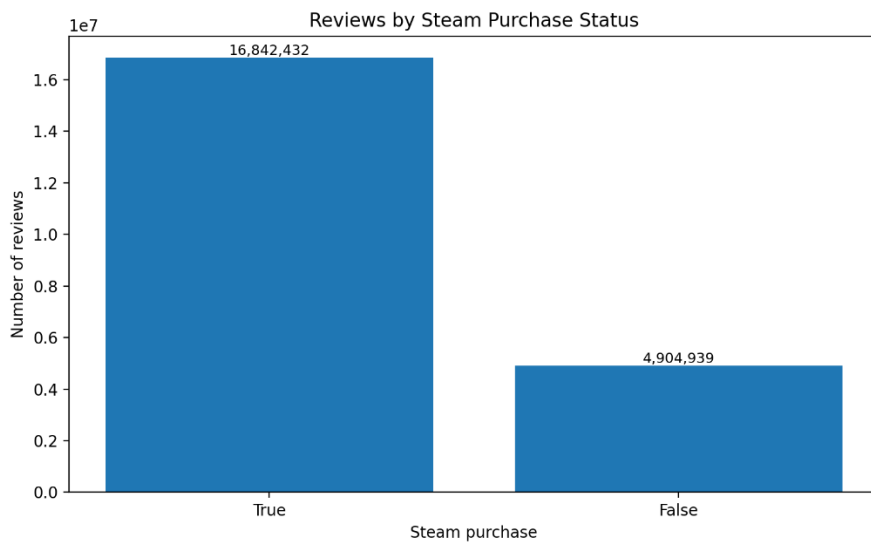


Figure 5 shows the relationship between recommendation status and free copies received by users. The majority of reviews were written by users who did not receive the game for free, indicating that most feedback came from paying customers.

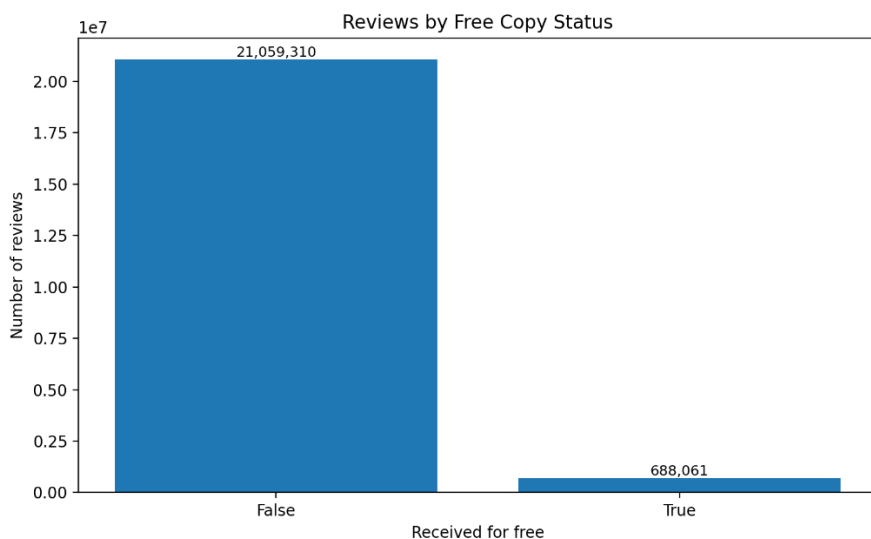


Figure 6 presents the average number of helpful votes for positive and negative reviews. Negative reviews received a higher average number of helpful votes compared to positive reviews, suggesting that critical reviews may contain more detailed information that other users find useful.

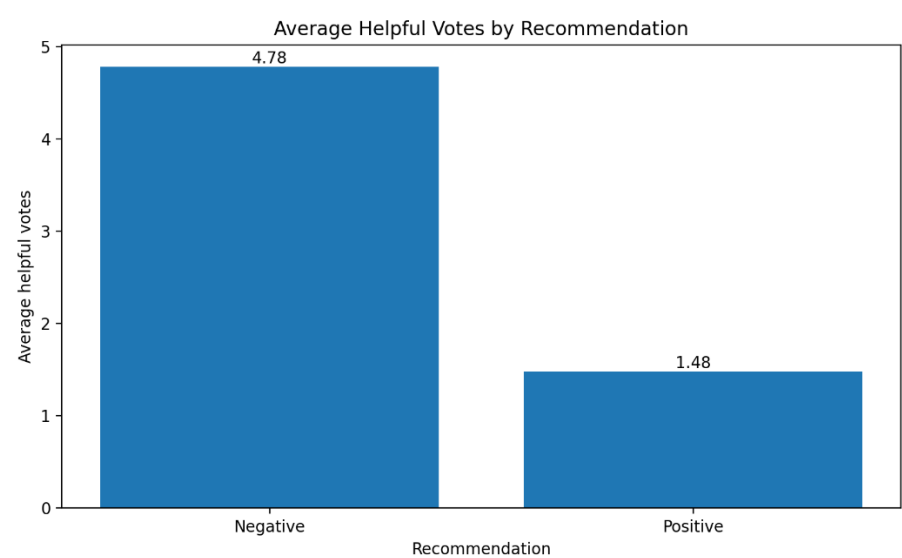
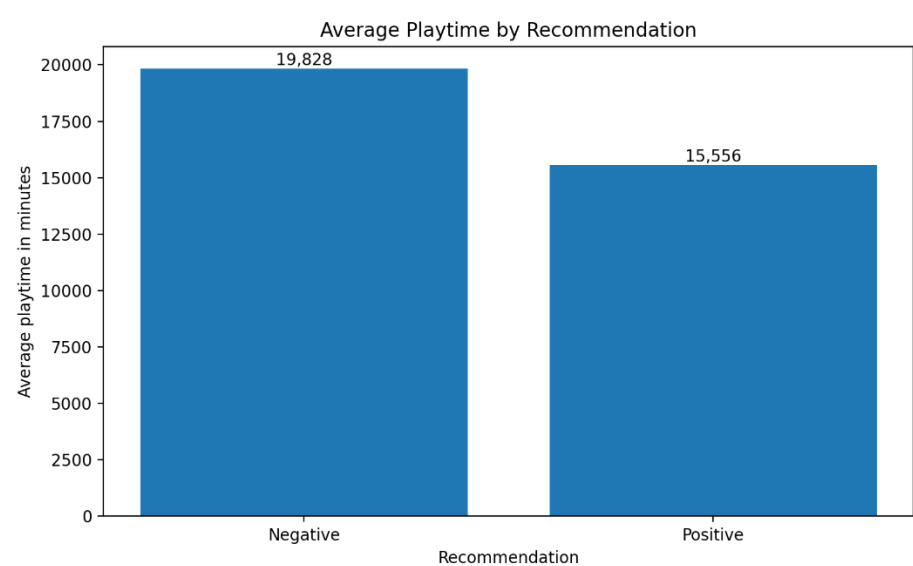


Figure 7 compares average playtime between positive and negative reviewers. Players who submitted negative reviews had, on average, more playtime than players who submitted positive reviews. This may indicate that experienced players are more likely to identify issues and provide critical feedback.



Recommendation Relationships

Category	Count
Steam purchase + Positive	14,731,705
Steam purchase + Negative	2,110,727
No Steam purchase + Positive	4,290,905
No Steam purchase + Negative	614,034
Not free + Positive	18,411,577
Not free + Negative	2,647,733
Free + Positive	611,033
Free + Negative	77,028

4. Discussion

The analysis revealed several interesting patterns within the Steam Reviews Dataset. Positive reviews dominated the dataset, indicating generally favorable user experiences across the analyzed games. The language distribution demonstrated the global reach of the Steam platform, with English serving as the primary communication language.

The popularity analysis showed that multiplayer and competitive games generated the largest amount of user engagement. Furthermore, negative reviews received a higher average number of helpful votes, suggesting that users often value detailed criticism when making purchasing decisions.

The playtime analysis revealed that users with negative reviews tended to spend more time playing before submitting their feedback. This observation may indicate that experienced players develop more critical perspectives after extensive gameplay. Overall, the findings demonstrate how large-scale gaming datasets can provide meaningful insights into player behavior and consumer preferences.

5. Conclusion

This project demonstrated the application of Hadoop and Spark technologies for large-scale data analytics using the Steam Reviews Dataset. The dataset was successfully stored in HDFS, processed using PySpark, and analyzed through various descriptive statistical methods. The results revealed important trends related to recommendation behavior, game popularity, language distribution, purchase patterns, and player engagement.

The project confirms that Big Data Analytics can be effectively applied to gaming datasets to uncover valuable patterns and support data-driven decision-making. Hadoop and Spark proved to be suitable technologies for handling large datasets and performing scalable analysis tasks.

6. References

Najzeko. *Steam Reviews Dataset 2021* [Internet]. Kaggle; 2021 [cited 2026 Jun 8]. Available from: <https://www.kaggle.com/datasets/najzeko/steam-reviews-2021>.

Apache Software Foundation. *Apache Hadoop Documentation* [Internet]. [cited 2026 Jun 8]. Available from: <https://hadoop.apache.org/docs/>.

Apache Software Foundation. *Apache Spark Documentation* [Internet]. [cited 2026 Jun 8]. Available from: <https://spark.apache.org/docs/latest/>.

7. Appendix

Passing Grade (G) Requirements	Implemented (Yes/No)	Comment
Req. 1: Data Collection	Yes	Steam Reviews Dataset (7.6 GB) downloaded from Kaggle and stored in HDFS
Req. 2: Data Preprocessing	Yes	Data cleaning, filtering, null handling and type conversion using PySpark
Req. 3: Smart Data Analysis	Yes	Aggregation, descriptive statistics and relationship analysis
Req. 4: Representation & Visualization	Yes	Bar charts and tables generated from Spark results
Req. 5: Security	Yes	Hadoop audit logging enabled for monitoring HDFS operations

